Scalable infrastructure for comparative genomics in the era of large-scale Tree of Life projects

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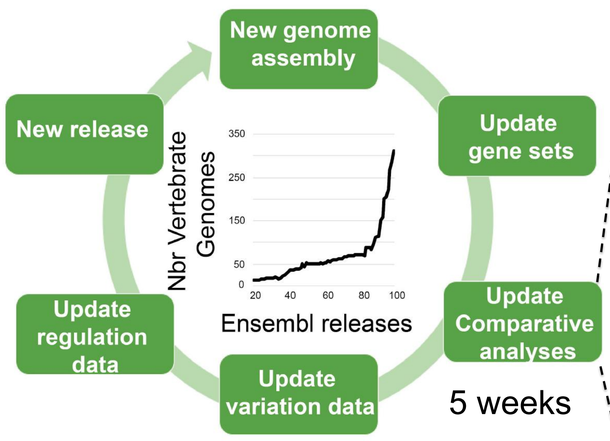
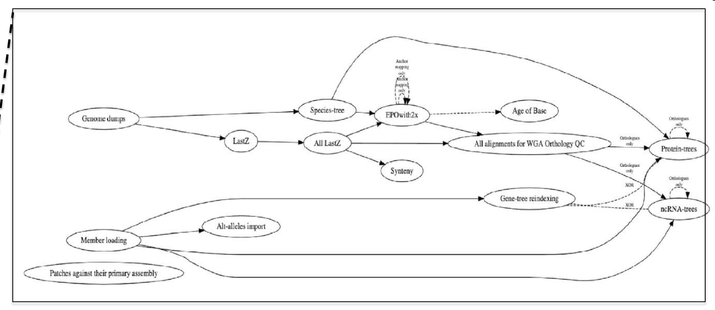
Introduction

Ensembl (www.ensembl.org) produces four releases per year. For each release cycle, integrated comparative analyses are computed for over 1,700 species. The exponential increase of genomes coupled to a five weeks time slot to produce the analyses required within Ensembl production cycle require continued improvement of efficiency of our algorithms, infrastructure and processes. Those improvements are becoming even more critical with the advent of ambitious large scale biodiversity sequencing project such as the Darwin tree of life project.





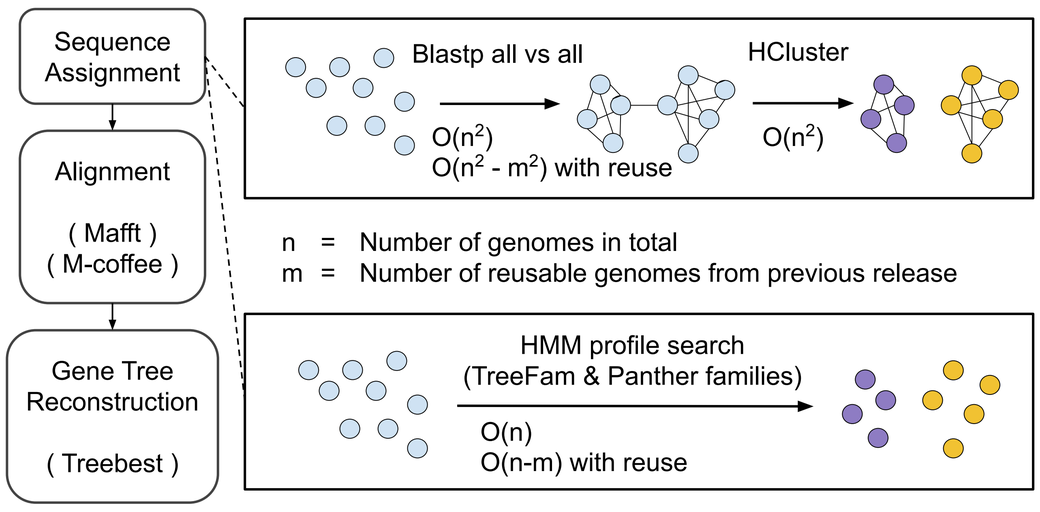
Set of interdependent comparative analyses that need to be computed in one release.

	e96	e100
Number of jobs (millions)	59.2	119.7
Average job time (s)	27.6	25.25
Total CPU years	51.8	96.19

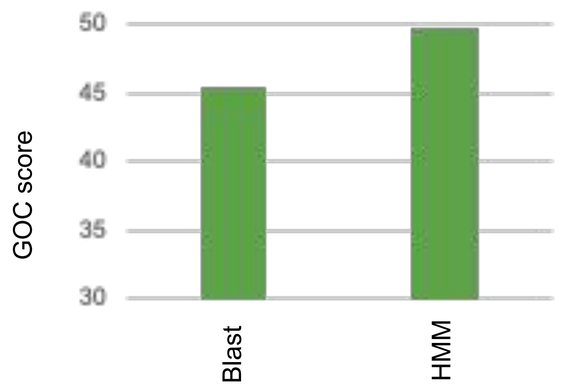
Our computing requirements have doubled in a year (between release 96 and release 100).

From quadratic to linear complexity methods

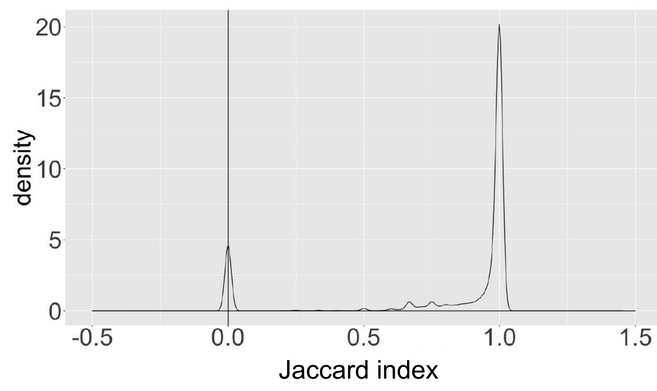
By switching to a Hidden Markov Model (HMM) classification to group sequences into families, we moved from a computation that grows quadratically with the number of species to a computation that grows linearly. This has also improved the specificity of our homology annotation.



Overview of the current gene-tree reconstruction workflow.



Gene order conservation (GOC) score measures the quality of orthology prediction based on the order conservation of neighbour genes. The HMM based approach has an overall better GOC score.



We used the Jaccard index to assess the similarity between our two approaches. Overall the HMM approach shows a good overlap with the Blast based clustering.

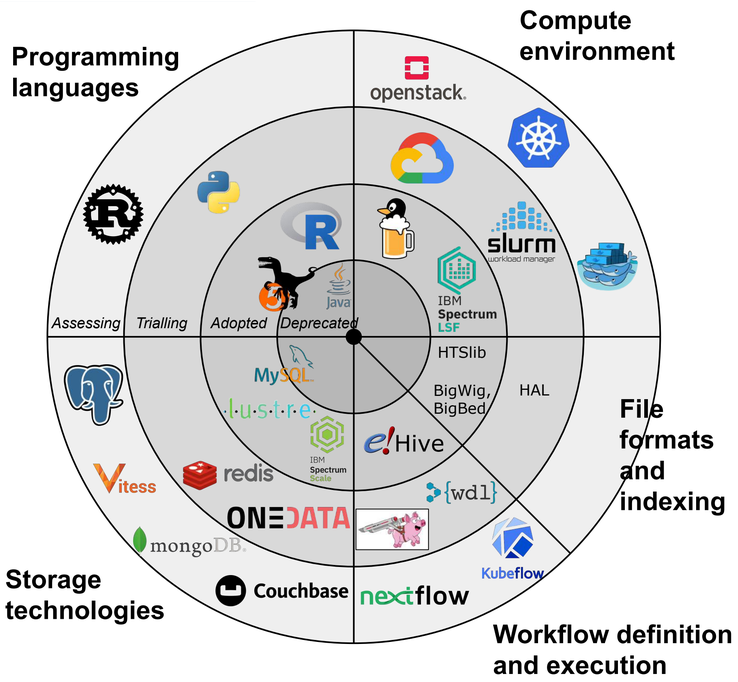
Infrastructure.

The amount of compute requires a fast and fail-proof system. The generation of the comparative analyses typically peaks at 6,000 concurrent jobs / 30 TB RAM, all managed by the eHive workflow manager, continuously improved by Ensembl since 2004.

eHive has a powerful graphical interface to display the workflow structure, monitor the execution, modify parameters on-the-fly, and manage failures, as well as detailed resource usage reports.

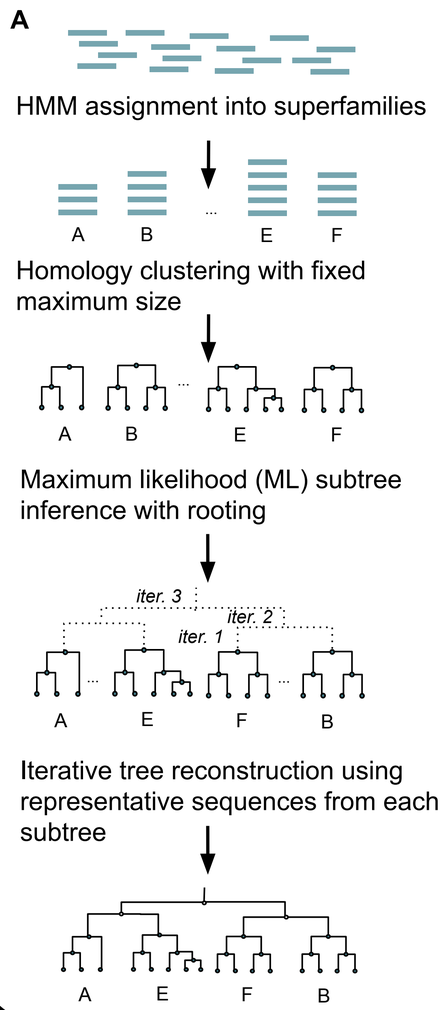
Ensembl has laid its foundations on Perl and MySQL, running the eHive workflows on a LSF cluster. We are continuously incorporating faster technologies into our infrastructure to scale further. For instance, regular updates to our GOC analysis over the last four years have made it 2,850 faster. Processing homologies off a Lustre file system instead of the MySQL database yielded a 5-fold speedup, saving 14 days of compute off the gene-tree reconstruction pipeline.

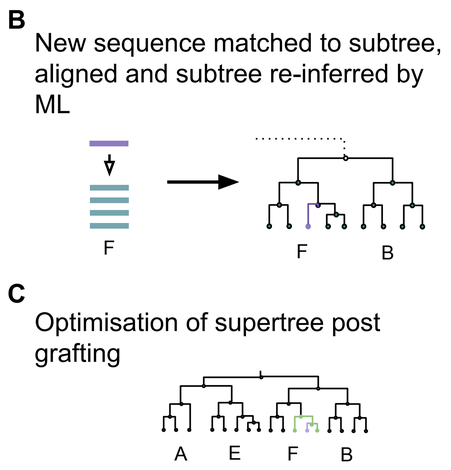
The *technology radar* on the right lists various alternatives we are exploring, grouped by topic, and arranged according to their adoption status (used in production, under trial, being assessed for future trial, or even deprecated).



Gene tree reconstruction at scale.

We are planning to move to a divide and conquer approach to reconstruct our gene trees .





***A** The initial release involves reconstructing gene trees for large homologous protein families.*
***B** The trees are updated iteratively every release, with some sequences removed and some inserted.*
***C** Finally, until the tree reaches a stable topology, there will be a periodic optimisation to ensure the most accurate orthology/paralogy inference possible.*

Whole genome alignments at scale.

Cactus is in development to be tuned it to our computer clusters and to be used as standard in our comparative analysis pipelines. Cactus provides the ability to iteratively add a genome to an existing alignment. The results will be compared to our current multiple WGA pipelines and eventually replace WGAs where improvement is detected.

Conclusion.

The improvement of the comparative analyses implemented by Ensembl over the time has enable the comparison of over 1700 genomes four times a year. We are now continuing further those improvements to enable the comparison of the 10s of thousands genomes expected to be produces by large scale genomic projects such as DToL

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